

R Pranav Ajay

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Education

Amrita School of Engineering

Bachelor of Technology Computer Science & Engineering *GPA: 8.04*

Ettimadai, Coimbatore

July 2018 - June 2022

Maharishi International Residential School Maharishi International Residential School

Sriperumbudur, Kanchipuram

Higher Secondary, CBSE Computer Science *GPA: XII- 455/500, X- 10 CGPA*

2016 - 2018

Experience

Caterpillar

Bengaluru, Karnataka, India

Senior Associate Engineer

August 2024 - Present

- Represented the Indian team on a business visit to Brazil, to analyze metrics & gather feedback.
- Worked on the deployment of a Project Chrono fork in the HIL(Hardware in Loop) & SIL (Software in Loop) environments.
- Worked on a Carla based fork application (Unreal Engine 4 Fork) & deployed it on HIL(Hardware in Loop) & SIL (Software in Loop) environments.
- Worked on profiling, optimization and overall graphical enhancement of modules within the project for the hardware environment.
- Worked on establishing CI/CD Pipelines end to end in the project scope.

Associate Engineer

August 2022 - August 2024

- Worked on a Carla fork & was involved on creating scenario generation, implementing a newer pipeline for the simulations
- Worked on a Project Chrono fork, implementing scene graphs, overall simulation pipeline, & asset handling pipeline.
- Worked on a SimPy fork, implementing scene graph wrappers for non-real-time simulations, & handling simulation logic.

College Intern

January 2022 - July 2022

- Worked on an Unreal Engine 4 fork & worked on custom scenario generation, asset pipeline, & scripting the simulations.
- Worked on a pipeline involving Unity & propriety tool chains & focused on creating scene graphs for the simulations.
- Worked on modeling the 3D assets & integrated them in the simulation pipeline.
- Having exposure to the C++ toolchains (Qt, CMake, vcpkg, msvc, clang) involved in the pipeline.

Skills

Programming Languages: C++, JavaScript, Python, GD Script

Frameworks: Unreal Engine, poco, boost, dear imgui, Godot, Blender, Three JS

Technical Interests: Graphics Rendering, Game Development, Computational Photography

Projects

Kreative Rum *Godot 4, Blender, Krita*

https://github.com/sairpa/kreative_rum

A Collection of all my projects pertaining to Game Development & pipeline

ToonLyt *TensorFlow Lite, Kotlin*

<https://github.com/sairpa/ToonLytV2>

An Android App with Pastiche creation abilities based on Google's pastiche tensorflow lite library

Content Creator *Kdenlive, Figma, Blender, YouTube*

<http://www.youtube.com/@palladiumos>

Created assets & animations for new releases & administrated the YouTube channel

User Interface Development *Figma, Android Studio, Kotlin*

<https://web.archive.org/web/20230306145421/https://palladiumos.com/>

Designed UI/UX prototypes & implemented them into the Android Apps